Remove Data Connections

We currently have a `team` table:

id	name	headquarters
1	Preventers	Sharp Tower
2	Z-Force	Sister Margaret's Bar

And a `hero` table:

id	name	secret_name	age	team_id
1	Deadpond	Dive Wilson	null	2
2	Rusty-Man	Tommy Sharp	48	1
3	Spider-Boy	Pedro Parqueador	null	1

Let's see how to **remove** connections between rows in tables.

We will continue with the code from the previous chapter.

👀 **Full file preview**

Python 3.10+

```python
from sqlmodel import Field, Session, SQLModel, create_engine


class Team(SQLModel, table=True):
    id: int | None = Field(default=None, primary_key=True)
    name: str = Field(index=True)
    headquarters: str


class Hero(SQLModel, table=True):
    id: int | None = Field(default=None, primary_key=True)
    name: str = Field(index=True)
    secret_name: str
    age: int | None = Field(default=None, index=True)

    team_id: int | None = Field(default=None, foreign_key="team.id")


sqlite_file_name = "database.db"
sqlite_url = f"sqlite:///{sqlite_file_name}"

engine = create_engine(sqlite_url, echo=True)


def create_db_and_tables():
    SQLModel.metadata.create_all(engine)


def create_heroes():
    with Session(engine) as session:
        team_preventers = Team(name="Preventers", headquarters="Sharp Tower")
        team_z_force = Team(name="Z-Force", headquarters="Sister Margaret's Bar")
        session.add(team_preventers)
        session.add(team_z_force)
        session.commit()

        hero_deadpond = Hero(
            name="Deadpond", secret_name="Dive Wilson", team_id=team_z_force.id
        )
        hero_rusty_man = Hero(
            name="Rusty-Man",
            secret_name="Tommy Sharp",
            age=48,
            team_id=team_preventers.id,
        )
        hero_spider_boy = Hero(name="Spider-Boy", secret_name="Pedro Parqueador")
        session.add(hero_deadpond)
        session.add(hero_rusty_man)
        session.add(hero_spider_boy)
        session.commit()

        session.refresh(hero_deadpond)
        session.refresh(hero_rusty_man)
        session.refresh(hero_spider_boy)

        print("Created hero:", hero_deadpond)
        print("Created hero:", hero_rusty_man)
        print("Created hero:", hero_spider_boy)

        hero_spider_boy.team_id = team_preventers.id
        session.add(hero_spider_boy)
        session.commit()
        session.refresh(hero_spider_boy)
        print("Updated hero:", hero_spider_boy)


def main():
    create_db_and_tables()
    create_heroes()
```

```
if __name__ == "__main__":
    main()
```

Break a Connection

We don't really have to delete anything to break a connection. We can just assign `None` to the foreign key, in this case, to the `team_id`.

Let's say **Spider-Boy** is tired of the lack of friendly neighbors and wants to get out of the **Preventers**.

We can simply set the `team_id` to `None`, and now it doesn't have a connection with the team:

Python 3.10+

```
# Code above omitted 👆

def create_heroes():
    with Session(engine) as session:

# Code here omitted 👈

        hero_spider_boy.team_id = None
        session.add(hero_spider_boy)
        session.commit()
        session.refresh(hero_spider_boy)
        print("No longer Preventer:", hero_spider_boy)

# Code below omitted 👇
```

👀 **Full file preview**

Python 3.10+

```python
from sqlmodel import Field, Session, SQLModel, create_engine


class Team(SQLModel, table=True):
    id: int | None = Field(default=None, primary_key=True)
    name: str = Field(index=True)
    headquarters: str


class Hero(SQLModel, table=True):
    id: int | None = Field(default=None, primary_key=True)
    name: str = Field(index=True)
    secret_name: str
    age: int | None = Field(default=None, index=True)

    team_id: int | None = Field(default=None, foreign_key="team.id")


sqlite_file_name = "database.db"
sqlite_url = f"sqlite:///{sqlite_file_name}"

engine = create_engine(sqlite_url, echo=True)


def create_db_and_tables():
    SQLModel.metadata.create_all(engine)


def create_heroes():
    with Session(engine) as session:
        team_preventers = Team(name="Preventers", headquarters="Sharp Tower")
        team_z_force = Team(name="Z-Force", headquarters="Sister Margaret's Bar")
        session.add(team_preventers)
        session.add(team_z_force)
        session.commit()

        hero_deadpond = Hero(
            name="Deadpond", secret_name="Dive Wilson", team_id=team_z_force.id
        )
        hero_rusty_man = Hero(
            name="Rusty-Man",
            secret_name="Tommy Sharp",
            age=48,
            team_id=team_preventers.id,
        )
        hero_spider_boy = Hero(name="Spider-Boy", secret_name="Pedro Parqueador")
        session.add(hero_deadpond)
        session.add(hero_rusty_man)
        session.add(hero_spider_boy)
        session.commit()

        session.refresh(hero_deadpond)
        session.refresh(hero_rusty_man)
        session.refresh(hero_spider_boy)

        print("Created hero:", hero_deadpond)
        print("Created hero:", hero_rusty_man)
        print("Created hero:", hero_spider_boy)

        hero_spider_boy.team_id = team_preventers.id
        session.add(hero_spider_boy)
        session.commit()
        session.refresh(hero_spider_boy)
        print("Updated hero:", hero_spider_boy)

        hero_spider_boy.team_id = None
        session.add(hero_spider_boy)
        session.commit()
        session.refresh(hero_spider_boy)
        print("No longer Preventer:", hero_spider_boy)
```

```python
def main():
    create_db_and_tables()
    create_heroes()


if __name__ == "__main__":
    main()
```

Again, we just **assign** a value to that field attribute `team_id`, now the value is `None`, which means `NULL` in the database. Then we `add()` the hero to the session, and then `commit()`.

Next we `refresh()` it to get the recent data, and we print it.

Running that in the command line will output:

```bash
```

That's it, we now removed a connection between rows in different tables by unsetting the foreign key column. 💥